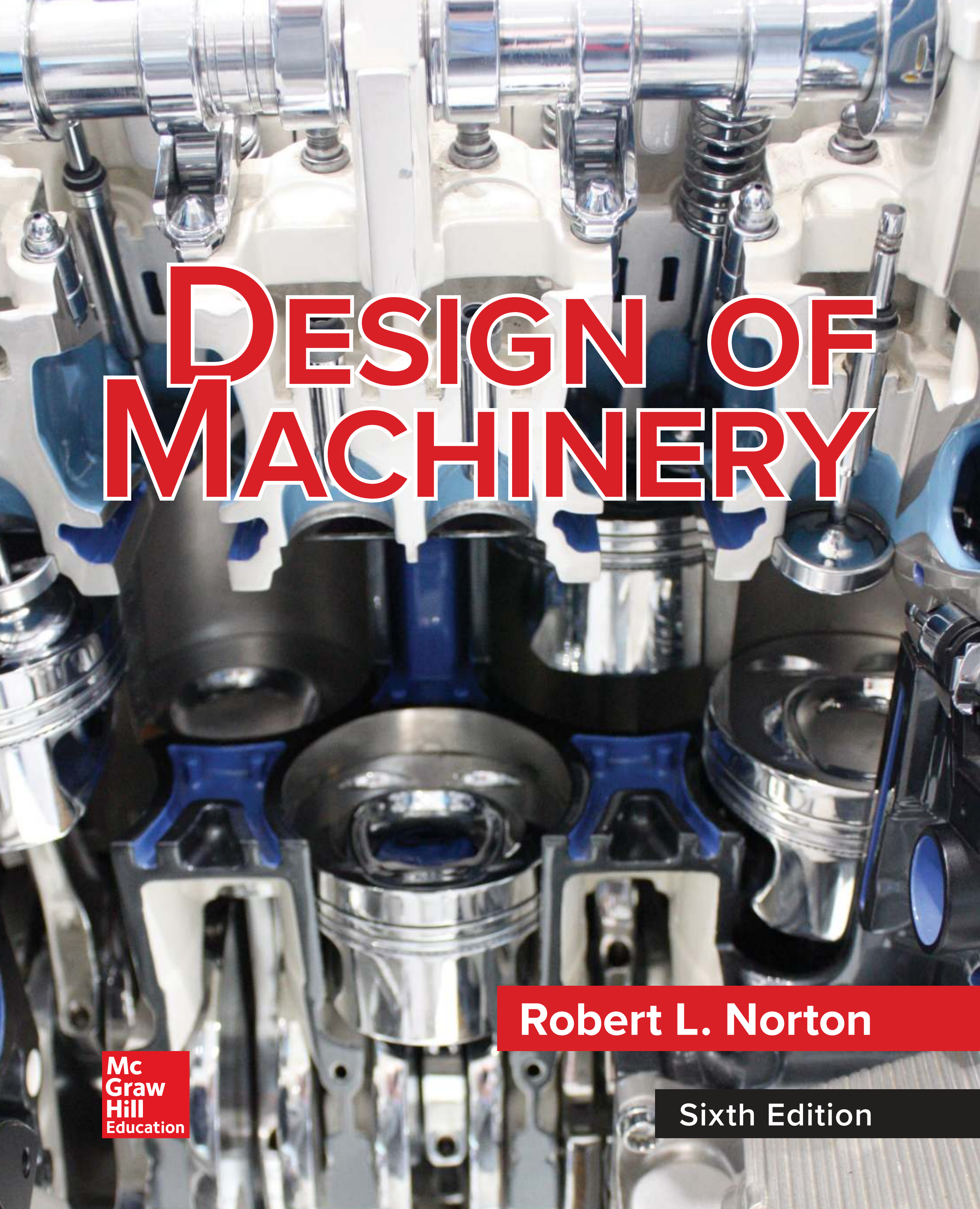


DESIGN OF MACHINERY

Robert L. Norton

Sixth Edition

McGraw Hill Education

DESIGN OF MACHINERY

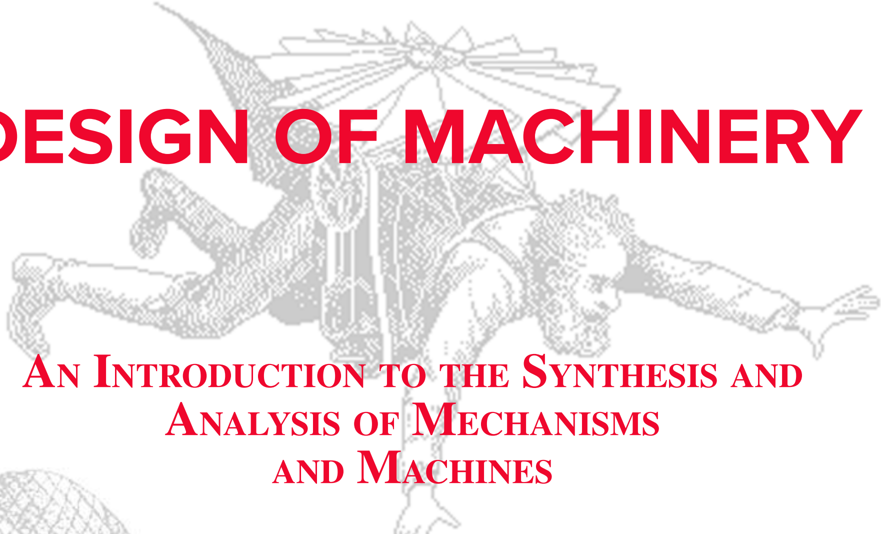
An Introduction to the Synthesis and Analysis of Mechanisms and Machines

Sixth Edition

McGraw Hill Education

McGraw-Hill Series in Mechanical Engineering

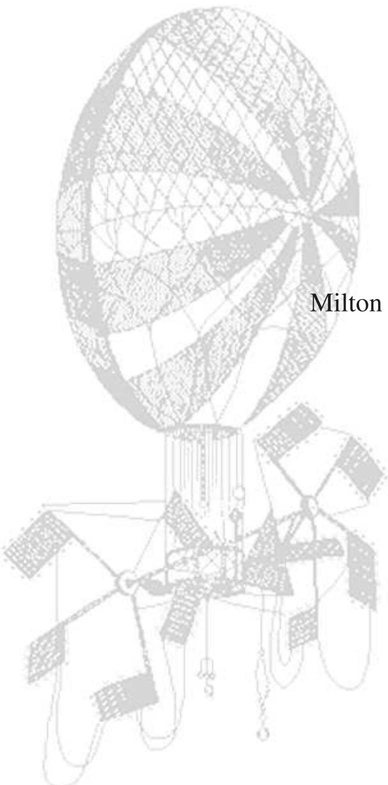
Alciatore	*Introduction to Mechatronics and Measurement Systems*
Anderson	*Fundamentals of Aerodynamics*
Anderson	*Introduction to Flight*
Anderson	*Modern Compressible Flow*
Beer/Johnston	*Vector Mechanics for Engineers*
Beer/Johnston	*Mechanics of Materials*
Budynas	*Advanced Strength and Applied Stress Analysis*
Budynas/Nisbett	*Shigley's Mechanical Engineering Design*
Byers/Dorf/Nelson	*Technology Ventures: From Idea to Enterprise*
Cengel	*Introduction to Thermodynamics and Heat Transfer*
Cengel/Boles	*Thermodynamics: An Engineering Approach*
Cengel/Cimbala	*Fluid Mechanics: Fundamentals and Applications*
Cengel/Ghajar	*Heat and Mass Transfer: Fundamentals and Applications*
Cengel/Turner/Cimbala	*Fundamentals of Thermal-Fluid Sciences*
Dieter/Schmidt	*Engineering Design*
Finnemore/Franzini	*Fluid Mechanics with Engineering Applications*
Heywood	*Internal Combustion Engine Fundamentals*
Holman	*Experimental Methods for Engineers*
Holman	*Heat Transfer*
Kays/Crawford/Weigand	*Convective Heat and Mass Transfer*
Norton	*Design of Machinery*
Palm	*System Dynamics*
Plesha/Gray/Costanzo	*Engineering Mechanics: Statics and Dynamics*
Reddy	*An Introduction to the Finite Element Method*
Schey	*Introduction to Manufacturing Processes*
Smith/Hashemi	*Foundations of Materials Science and Engineering*
Turns	*An Introduction to Combustion: Concepts and Applications*
Ullman	*The Mechanical Design Process*
White	*Fluid Mechanics*
White	*Viscous Fluid Flow*
Zeid	*Mastering CAD/CAM*



DESIGN OF MACHINERY

An Introduction to the Synthesis and Analysis of Mechanisms and Machines

Sixth Edition

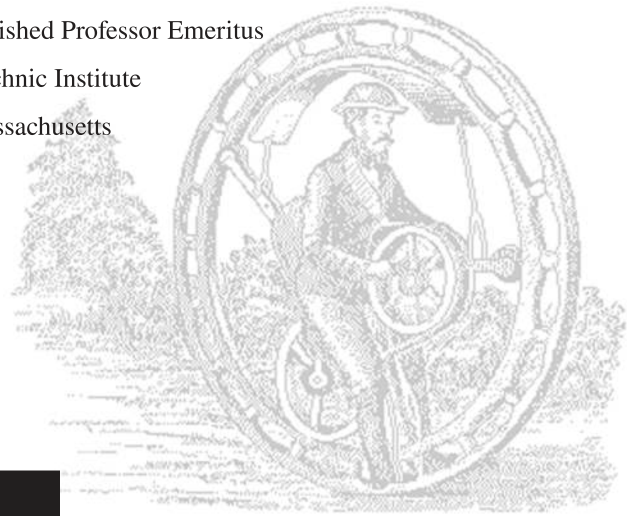


Robert L. Norton

Milton P. Higgins II Distinguished Professor Emeritus

Worcester Polytechnic Institute

Worcester, Massachusetts



Mc Graw Hill Education

Design of Machinery: An Introduction to the Synthesis and Analysis of Mechanisms and Machines, Sixth Edition

Published by McGraw-Hill Education, 2 Penn Plaza, New York, NY, 10121.

Some ancillaries, including electronic and print components, may not be available to customers outside the United States.

This book is printed on acid-free paper containing 10% postconsumer waste.

1 2 3 4 5 6 7 8 9 LCR 21 20 19

ISBN 978-1-260-11331-0 (bound edition)
MHID 1-260-11331-0 (bound edition)
ISBN 978-1-260-43130-8X (loose-leaf edition)
MHID 1-260-43130-4 (loose-leaf edition

Portfolio Manager: Thomas Scaife, Ph.D.
Product Developers: Tina Bower; Megan Platt
Marketing Manager: Shannon O'Donnell
Content Project Managers: Jeni McAtee; Rachael Hillebrand
Buyer: Susan K. Culbertson
Design: Debra Kubiak
Content Licensing Specialist: Melissa Homer
Cover Image and Design: ©Robert L. Norton
Compositor: Robert L. Norton and Aptara, Inc.

All credits appearing on page or at the end of the book are considered to be an extension of the copyright page.

All text, drawings, and equations in this print book were prepared and typeset electronically, by the author, on a *Macintosh*® computer using *Illustrator*®, *MathType*®, and *InDesign*® desktop publishing software. The body text is set in Stix General, and headings set in Proxima Nova. Color-separated printing plates were made on a direct-to-plate laser typesetter from the author's files. All *clip art* illustrations are courtesy of *Dubl-Click Software Inc.*, 22521 Styles St., Woodland Hills CA 91367, reprinted from their *Industrial Revolution* and *Old Earth Almanac* series with their permission (and with the author's thanks). The cover photo, by the author, is a cut-away of a VR-15 six-cylinder engine.

Library of Congress Cataloging-in-Publication Data

Names: Norton, Robert L., Author

Title: Design of machinery : an introduction to the synthesis and analysis of mechanisms and machines / Robert L. Norton.

Description: Sixth edition. | New York, NY : McGraw-Hill, [2020] | Includes bibliographical references and index.

Identifiers: LCCN 2018034424| ISBN 9781260113310 (alk. paper) | ISBN 1260113310 (alk. paper)

Subjects: LCSH: Machine design. | Machinery, Kinematics of. | Machinery, Dynamics of.

Classification: LCC TJ230 .N63 2020 | DDC 621.8/15--dc23 LC record available at https://lccn.loc.gov/2018034424

Publisher's web page is mheducation.com/highered. Author's web page is http://www.designofmachinery.com

ABOUT THE AUTHOR

Robert L. Norton earned undergraduate degrees in both mechanical engineering and industrial technology at Northeastern University and an MS in engineering design at Tufts University. He was awarded a Dr. Eng. (h.c.) by WPI in 2012. He is a registered professional engineer in Massachusetts and Florida. He has extensive industrial experience in engineering design and manufacturing and many years' experience teaching mechanical engineering, engineering design, computer science, and related subjects at Northeastern University, Tufts University, and Worcester Polytechnic Institute.

At Polaroid Corporation for 10 years, he designed cameras, related mechanisms, and high-speed automated machinery. He spent three years at Jet Spray Cooler Inc., designing food-handling machinery and products. For five years he helped develop artificial-heart and noninvasive assisted-circulation (counterpulsation) devices at the Tufts New England Medical Center and Boston City Hospital. Since leaving industry to join academia, he has continued as an independent consultant on engineering projects ranging from disposable medical products to high-speed production machinery. He holds 13 U.S. patents.

Norton has been on the faculty of Worcester Polytechnic Institute since 1981 and is currently Milton Prince Higgins II Distinguished Professor Emeritus in the mechanical engineering department. He taught undergraduate and graduate courses in mechanical engineering with an emphasis on design, kinematics, vibrations, and dynamics of machinery for 31 years at WPI before retiring from active teaching.

He is the author of numerous technical papers and journal articles covering kinematics, dynamics of machinery, cam design and manufacturing, computers in education, and engineering education and of the texts *Machine Design: An Integrated Approach, 6ed* and the *Cam Design and Manufacturing Handbook, 2ed*. He is a Fellow and life member of the American Society of Mechanical Engineers and a past member of the Society of Automotive Engineers. In 2007, he was chosen as *U. S. Professor of the Year* for the State of Massachusetts by the *Council for the Advancement and Support of Education (CASE)* and the *Carnegie Foundation for the Advancement of Teaching*, who jointly present the only national awards for teaching excellence given in the United States of America.

WHAT USERS SAY ABOUT THE BOOK

Your text is the best of all the texts I have used—the balance of fundamentals and practice is especially important, and you have achieved that with aplomb!

—Professor John P. H. Steele, *Colorado School of Mines*

We picked your book (years ago) because it was (and still is) the most accessible undergraduate presentation of the material I have found; although clearly theoretically based (and not afraid of the math!), you have a style and method that brings the material alive to students.

—Professor Michael Keefe, *University of Delaware*

As an instructor who has been using Design of Machinery in my classes for over 12 years, I've been especially impressed throughout by Professor Norton's care and attention in support of its users.

—Professor John Lee, *San Diego State University*

This book is dedicated to the memory of my father,

Harry J. Norton, Sr.

who sparked a young boy's interest in engineering;

to the memory of my mother,

Kathryn T. Norton

who made it all possible;

to my wife,

Nancy Norton

who provides unflagging patience and support;

and to my children,

Robert, Mary, and Thomas,

who make it all worthwhile.

PREFACE
to the Sixth Edition

The sixth edition is an evolutionary improvement over the fifth and earlier editions. See the updated *Preface to the First Edition* (overleaf) for more detailed information on the book's purpose and organization. The principal changes in this edition are:

- In addition to the printed version of the text, digital e-book versions are also available. These have hotlinks to all the videos and to the downloadable content provided. There are 188 videos. All of these are marked in the print version as well, with their URLs provided, and they can be downloaded by print-book users. A **Video Contents** is provided, and all other downloadable items are listed in the **Downloads Index**.

- Over 50 new problem assignments have been added. The problem figures are included as downloadable PDF files so that students can easily print hard copies on which to work the solutions.

- The author-written programs that come with the book have been completely rewritten to improve their interface and usability, and they are now compatible with the latest operating systems and computers. The programs FOURBAR, FIVEBAR, SIXBAR, SLIDER, and ENGINE have been combined in a new program called LINKAGES that does everything those programs collectively did with new features added. Program DYNACAM also has been completely rewritten and is much improved. Program MATRIX is updated. These computer programs undergo frequent revision to add features and enhancements. Professors who adopt the book for a course and students using the print book may register to download the latest student versions of these programs from: http://www.designofmachinery.com. Click on the *Student* or *Professor* link.

- The *Working Model* program is needed to run the Working Model files included with this text. Some universities have site licenses for this program on their lab computers. The supplier, *Design Simulation Technologies,* offers student licenses for one-semester or one-year periods at moderate cost. These are available at http://www.design-simulation.com/Purchase/studentproducts.php.

- Many small improvements have been made to the discussion of a variety of topics in many chapters, based largely on user feedback, and all known errors have been corrected.

The extensive DVD content that was introduced in the Fifth Edition is now downloadable from a website. These downloads include:

- The entire *Hrones and Nelson Atlas of Coupler Curves* and the *Zhang et al Atlas of Geared Fivebar Coupler Curves*.

- Wang's *Mechanism Simulation in a Multimedia Environment* contains 105 Working Model (WM) files based on the book's figures with AVI files and 19 Matlab® models for kinematic analysis and animation. The AVI files are linked to their figures in the e-books.

- Videos of two "virtual laboratories" that replicate labs created by the author at WPI are provided. These include demonstrations of the lab machines used and spreadsheet files of the acceleration and force data taken during the experiments. The intent is to allow students at other schools to do these exercises as virtual laboratories.

- A series of 34 *Master Lecture Videos* by the author that cover most of the topics in the book as well as 39 shorter "snippets" from these lectures are woven into the chapters. Seven *Demonstration Videos* are also provided. These were recorded over the author's thirty-one years of teaching these subjects at WPI and are listed in the **Video Contents**.

All the downloadable files are accessible to digital-book users through the publisher's website via links in the digital book. Any instructor or student who uses the print book may register on my website, *http://www.designofmachinery.com*, either as a student or instructor, and I will send them a password to access a protected site where they can download the latest versions of my computer programs, LINKAGES, DYNACAM, and MATRIX, all videos, and all files listed in the Downloads Index. Note that I personally review each of these requests for access and approve only those that are filled out completely and correctly according to the provided instructions. I require complete information and only accept university email addresses.

ACKNOWLEDGMENTS The sources of photographs and other nonoriginal art used in the text are acknowledged in the captions and opposite the title page, but the author would also like to express his thanks for the cooperation of all those individuals and companies who generously made these items available. The author is indebted to, and would like to thank, a number of users who kindly notified him of errors or suggested improvements in all editions since the first. These include: Professors *Chad O'Neal* of Louisiana Tech, *Bram Demeulenaere* of Leuven University, *Eben Cobb* of WPI, *Diego Galuzzi* of University of Buenos Aires, *John R. Hall* of WPI, *Shafik Iskander* of U. Tennessee, *Richard Jakubek* of RPI, *Cheong Gill-Jeong* of Wonkwang University, Korea, *Swami Karunamoorthy* of St. Louis University, *Pierre Larochelle* of Florida Tech, *Scott Openshaw* of Iowa State, *Francis H. Raven* of Notre Dame, *Arnold E. Sikkema* of Dordt College, and *Donald A. Smith* of U. Wyoming.

Professors *M. R. Corley* of Louisiana Tech, *R. Devashier* of U. Evansville, *K. Gupta* of U. Illinois-Chicago, *M. Keefe* of U. Delaware, *J. Steffen* of Valparaiso University, *D. Walcerz* of York College, and *L. Wells* of U. Texas at Tyler provided useful suggestions or corrections. Professors *L. L. Howell* of BYU, and *G. K. Ananthasuresh* of U. Penn supplied photographs of compliant mechanisms. Professor *C. Furlong* of WPI generously supplied MEMS photos and information. Special thanks to *James Cormier* and *David Taranto* of WPI's Academic Technology Center for their help in creating the videos.

The author would like to express his appreciation to Professor *Sid Wang* of NCAT and his students for their efforts in creating the *Working Model* and *Matlab* files. Professor *Thomas A. Cook*, Mercer University (Emeritus), provided most of the new problem sets as well as their solutions in his impressive and voluminous solutions manual and its accompanying *Mathcad*® solution files. The author is most grateful for Dr. Cook's valuable contributions. Most of all, the author again thanks his infinitely patient wife, Nancy, who has provided unflagging support to him in all his endeavors for the past fifty-eight years.

If you find any errors or have comments or suggestions for improvement, please email the author at norton@wpi.edu. Errata as discovered, and other book information, will be posted on the author's web site at *http://www.designofmachinery.com*.

Robert L. Norton
Mattapoisett, Mass.
August, 2018

PREFACE
to the First Edition

When I hear, I forget
When I see, I remember
When I do, I understand
ANCIENT CHINESE PROVERB

This text is intended for the kinematics and dynamics of machinery topics which are often given as a single course, or two-course sequence, in the junior year of most mechanical engineering programs. The usual prerequisites are first courses in statics, dynamics, and calculus. Usually, the first semester, or portion, is devoted to kinematics and the second to dynamics of machinery. These courses are ideal vehicles for introducing the mechanical engineering student to the process of design, since mechanisms tend to be intuitive for the typical mechanical engineering student to visualize and create.

While this text attempts to be thorough and complete on the topics of analysis, it also emphasizes the synthesis and design aspects of the subject to a greater degree than most texts in print on these subjects. Also, it emphasizes the use of computer-aided engineering as an approach to the design and analysis of this class of problems by providing software that can enhance student understanding. While the mathematical level of this text is aimed at second- or third-year university students, it is presented *de novo* and should be understandable to the technical school student as well.

Part I of this text is suitable for a one-semester or one-term course in kinematics. Part II is suitable for a one-semester or one-term course in dynamics of machinery. Alternatively, both topic areas can be covered in one semester with less emphasis on some of the topics covered in the text.

The writing and style of presentation in the text are designed to be clear, informal, and easy to read. Many example problems and solution techniques are presented and spelled out in detail, both verbally and graphically. All the illustrations are done with computer-drawing or drafting programs. Some scanned photographic images are also included. The entire text, including equations and artwork, is printed directly from the author's PDF files by laser typesetting for maximum clarity and quality. Many suggested readings are provided in the bibliography. Short problems and, where appropriate, many longer, unstructured design project assignments are provided at the ends of chapters. These projects provide an opportunity for the students *to do and understand.*

The author's approach to these courses and this text is based on over 40 years' experience in mechanical engineering design, both in industry and as a consultant. He has taught these subjects since 1967, both in evening school to practicing engineers and in day school to younger students. His approach to the course has evolved a great deal in that time, from a traditional approach, emphasizing graphical analysis of many structured problems, through emphasis on algebraic methods as computers became available, through requiring students to write their own computer programs, to the current state described above.

The one constant throughout has been the attempt to convey the art of the design process to the students in order to prepare them to cope with *real* engineering problems in practice. Thus, the author has always promoted design within these courses. Only recently, however, has technology provided a means to more effectively accomplish this goal, in the form of the graphics microcomputer. This text attempts to be an improvement over those currently available by providing up-to-date methods and techniques for analysis and synthesis that take full advantage of the graphics microcomputer, and by emphasizing design as well as analysis. The text also provides a more complete, modern, and thorough treatment of cam design than any existing texts in print on the subject.

The author has written three interactive, student-friendly computer programs for the design and analysis of mechanisms and machines. These programs are designed to enhance the student's understanding of the basic concepts in these courses while simultaneously allowing more comprehensive and realistic problem and project assignments to be done in the limited time available than could ever be done with manual solution techniques, whether graphical or algebraic. Unstructured, realistic design problems which have many valid solutions are assigned. Synthesis and analysis are emphasized equally. The analysis methods presented are up to date, using vector equations and matrix techniques wherever applicable. Manual graphical analysis methods are deemphasized. The graphics output from the computer programs allows the student to see the results of variation of parameters rapidly and accurately and reinforces learning.

These computer programs are distributed with this book, and can be run on any Windows NT/2000/XP/Vista/Windows7/8/10 capable computer. Program LINKAGES analyzes the kinematics and dynamics of fourbar, geared fivebar, sixbar, and fourbar slider linkages. It also will synthesize fourbar linkages for two and three positions. LINKAGES also analyzes the slider-crank linkage as used in the internal combustion engine and provides a complete dynamic analysis of single- and multicylinder engine inline, V, and W configurations, allowing the mechanical dynamic design of engines to be done. DYNACAM allows the design and dynamic analysis of cam-follower systems. MATRIX is a general-purpose linear equation system solver. These are student editions of professional programs that are written by the author and that he provides to companies the world over.

All these programs, except MATRIX, provide dynamic, graphical animation of the designed devices. The reader is strongly urged to make use of these programs in order to investigate the results of variation of parameters in these kinematic devices. The programs are designed to enhance and augment the text rather than be a substitute for it. The converse is also true. Many solutions to the book's examples and to the problem sets are downloadable as files to be opened in these programs. Most of these solutions can be animated on the computer screen for a better demonstration of the concept than is possible on the printed page. The instructor and students are both encouraged to take advantage of

The author's intention is that synthesis topics be introduced first to allow the students to work on some simple design tasks early in the term while still mastering the analysis topics. Though this is not the "traditional" approach to the teaching of this material, the author believes that it is a superior method to that of initial concentration on detailed analysis of mechanisms for which the student has no concept of origin or purpose.

Chapters 1 and 2 are introductory. Those instructors wishing to teach analysis before synthesis can leave Chapters 3 and 5 on linkage synthesis for later consumption. Chapters 4, 6, and 7 on position, velocity, and acceleration analysis are sequential and build upon each other. In fact, some of the problem sets are common among these three chapters so that students can use their position solutions to find velocities and then later use both to find the accelerations in the same linkages. Chapter 8 on cams is more extensive and complete than that of other kinematics texts and takes a design approach. Chapter 9 on gear trains is introductory. The dynamic force treatment in Part II uses matrix methods for the solution of the system simultaneous equations. Graphical force analysis is not emphasized. Chapter 10 presents an introduction to dynamic systems modeling. Chapter 11 deals with force analysis of linkages. Balancing of rotating machinery and linkages is covered in Chapter 12. Chapters 13 and 14 use the internal combustion engine as an example to pull together many dynamic concepts in a design context. Chapter 15 presents an introduction to dynamic systems modeling and uses the cam-follower system as the example. Chapter 16 describes servo- and cam-driven linkages. Chapters 3, 8, 11, 13, and 14 provide open-ended project problems as well as structured problem sets. The assignment and execution of unstructured project problems can greatly enhance the student's understanding of the concepts as described by the proverb in the epigraph to this preface.

ACKNOWLEDGMENTS The sources of photographs and other nonoriginal art used in the text are acknowledged in the captions and opposite the title page, but the author would also like to express his thanks for the cooperation of all those individuals and companies who generously made these items available. The author would also like to thank those who reviewed various sections of the first edition of the text and who made many useful suggestions for improvement. Mr. John Titus of the University of Minnesota reviewed Chapter 5 on analytical synthesis and Mr. Dennis Klipp of Klipp Engineering, Waterville, Maine, reviewed Chapter 8 on cam design. Professor William J. Crochetiere and Mr. Homer Eckhardt of Tufts University, Medford, MA., reviewed Chapter 15. Mr. Eckhardt and Professor Crochetiere of Tufts, and Professor Charles Warren of the University of Alabama taught from and reviewed Part I. Professor Holly K. Ault of Worcester Polytechnic Institute thoroughly reviewed the entire text while teaching from the prepublication, class-test versions of the complete book. Professor Michael Keefe of the University of Delaware provided many helpful comments. Sincere thanks also go to the large number of undergraduate students and graduate teaching assistants who caught many typos and errors in the text and in the programs while using prepublication versions. Since the book's first printing, Profs. D. Cronin, K. Gupta and P. Jensen and Mr. R. Jantz have written to point out errors or make suggestions that I have incorporated and for which I thank them. The author takes full responsibility for any errors that may remain and invites from all readers their criticisms, suggestions for improvement, and identification of errors in the text or programs, so that both can be improved in future versions. Contact norton@wpi.edu.

Robert L. Norton
Mattapoisett, Mass.
August, 1991

CONTENTS

VIDEO CONTENTS

The Sixth Edition has a collection of *Master Lecture Videos and Tutorials* made by the author over a thirty-year period while teaching at Worcester Polytechnic Institute. The lectures were recorded in a classroom in front of students in 2011/2012. Tutorials were done in a recording studio and were intended as supplements to class lectures. Some have become lectures here.

There are 82 instructional videos in total. Thirty-four are "50-minute" lectures and are labeled as such. Thirty-nine are short "snippets" from the lecture videos that are linked to the relevant topics in a chapter. Seven are demonstrations of machinery or tutorials. Two are laboratory excercises that have been "virtualized" via video demonstration and the provision of test data so that students can simulate the lab. The run times of all videos are noted in the tables.

The sixth edition is available both as a print book and as digital media. The digital, e-book versions have active links that allow these videos to be run while reading the book. The print edition notes the names and URLs of all the videos in the text at their links.

In addition to the lecture videos, all the digital content that was with the fifth and earlier editions is still available as downloads, including the author-written programs LINKAGES, DYNACAM, and MATRIX. An index of all these files is in the **Downloads Index** and includes 105 video animations of figures in the book. In the digital e-book versions, these are hotlinked to the figures that they animate. The URL of each figure video is also provided in the figure for print-book readers to download them.

Because this book is about the study of motion, it is particularly well suited to digital media. The figure animations have been in the book since the third edition as files on a DVD. It is unknown how many students bothered to open and run those files. The digital versions of the book will make this much easier to do.

Any instructor or student who uses the print book may register on my website, http://www.designofmachinery.com, either as a student or instructor, and I will send them a password to access a protected site where they can download the latest versions of my computer programs, LINKAGES, DYNACAM, and MATRIX. They can also view the 82 videos and download all the files listed in the Download Index. Note that I personally review each of these requests for access and will approve only those that are filled out completely and correctly according to the instructions. I require complete information and only accept university email addresses.

LECTURE VIDEOS AND SNIPPETS Part 1 *(Concatenate this URL with any filename below to run a video)*

Chapter	Lecture	Snippet	Topic	http://www.designofmachinery.com/DOM/	Run Time
1	**1**		**Introduction**	**Introduction.mp4**	39:10
1	**2**		**Design Process and Documentation**	**Design_Process_and_Documentation.mp4**	43:28
1		2A	*Design Process*	*Design_Process.mp4*	29:46
1		2B	*Documentation*	*Documentation.mp4*	15:57
1		2C	*Units of Mass*	*Units.mp4*	10:06
2	**3**		**Kinematics Fundamentals**	**Kinematics_Fundamentals.mp4**	49:12
2		3A	*Degree of Freedom*	*DOF.mp4*	03:53
2		3B	*Links and Joints*	*Links_and_Joints.mp4*	11:00
2		3C	*Grubler*	*Links_and_Joints.mp4*	14:29
2		3D	*Number Synthesis*	*Number_Synthesis.mp4*	03:47

LECTURE VIDEOS AND SNIPPETS Part 2

(*Concatenate this URL with any filename below to run a video*)

Chapter	Lecture	Snippet	Topic	http://www.designofmachinery.com/DOM/	Run Time
2		3E	*Isomers*	*Isomers.mp4*	04:15
2		3F	*Inversion*	*Inversion.mp4*	03:40
2		3G	*Grashof Condition*	*The_Grashof_Condition.mp4*	07:21
2		3H	*Fivebar Linkage*	*Fivebar.mp4*	01:33
2		3I	*Compliant Linkages*	*Compliant_Linkages.mp4*	01:27
3	**4**		**Position Synthesis**	**Position_Synthesis.mp4**	47:57
3	**5**		**Quick Return Linkages**	**Quick_Return_Linkages.mp4**	55:22
3	**6**		**Coupler Curves and Atlases**	**Coupler_Curves.mp4**	59:57
3	**7**		**Symmetrical and Straight Coupler Curves**	**Symmetrical_and_Straight_Coupler_Curves.mp4**	14:47
3		7A	*Symmetrical Coupler Curves*	*Symmetrical_Coupler_Curves.mp4*	05:47
3		7B	*Straight Line Linkages*	*Straight_Line_Linkages.mp4*	09:20
3	**8**		**Cognates of Linkages**	**Cognates_of_Linkages.mp4**	18:12
3	**9**		**Parallel Motion**	**Parallel_Motion.mp4**	21:49
3	**10**		**Dwell Linkages**	**Dwell_Mechanisms.mp4**	35:36
4	**11**		**Position Analysis**	**Position_Analysis.mp4**	49:48
5	**12**		**Analytical Linkage Synthesis**	**Analytical_Linkage_Synthesis.mp4**	48:17
6	**13**		**Instant Centers and Centrodes**	**Instant_Centers_and_Centrodes.mp4**	49:16
6		13A	*Instant Centers*	*Instant_Centers_Tutorial.mp4*	28:55
6		13B	*Centrodes*	*Centrodes.mp4*	21:01
6	**14**		**Velocity Analysis with ICs**	**Velocity_Analysis_with_ICs.mp4**	49:48
6	**15**		**Velocity Analysis with Vectors**	**Velocity_Analysis_with_Vectors.mp4**	46:41
7	**16**		**Acceleration Analysis**	**Acceleration_Analysis.mp4**	41:39
7	**Lab**		**Fourbar Virtual Laboratory Video**	**Fourbar_Machine_Virtual_Laboratory.mp4**	35:38
7	**Lab**		**Fourbar Virtual Laboratory Data**	**Fourbar_Virtual_Lab.zip**	–
8	**17**		**Cam Design I**	**Cam_Design_I.mp4**	50:42
8	**18**		**Cam Design II**	**Cam_Design_II.mp4**	51:16
8	**19**		**Cam Design III**	**Cam_Design_III.mp4**	48:54
8	**Lab**		**Cam Machine Virtual Laboratory Video**	**Cam_Machine_Virtual_Laboratory.mp4**	21:28
8	**Lab**		**Cam Machine Laboratory Data**	**Cam_Virtual_Lab.zip**	–
9	**20**		**Gear Design**	**Gear_Design.mp4**	54:45
9	**21**		**GearTrains**	**Gear_Trains.mp4**	37:53
9	**22**		**Gear Transmissions**	**Gear_Transmissions.mp4**	41:06
10	**23**		**Dynamics Fundamentals**	**Dynamics_Fundamentals.mp4**	52.01
10		23A	*Newtons Laws*	*Newtons_Laws.mp4*	04:00
10		23B	*Units of Mass*	*Mass.mp4*	10:06
10		23C	*Moments of Mass*	*Moments_of_Mass.mp4*	04:33
10		23D	*Transfer Theorem*	*Transfer_Theorem.mp4*	02:15
10		23E	*Radius of Gyration*	*Radius_of_Gyration.mp4*	01:21
10		23F	*Rotating Links*	*Rotating_Links.mp4*	02:11
10		23G	*Center of Percussion*	*Center_of_Percussion.mp4*	06:58
10		23H	*Equivalent Systems*	*Equivalent_Systems.mp4*	14:53
10		23I	*Lumped Models*	*Lumped_Models.mp4*	19:39
10		23J	*Lever Ratios*	*Lever_Ratios.mp4*	05:16
10		23K	*Modeling Systems*	*Modeling_Systems.mp4*	06:44
10		23L	*DAlembert Force*	*DAlembert_Force.mp4*	03:36
10		23M	*Centrifugal Force*	*Centrifugal_Force.mp4*	06:58

LECTURE VIDEOS AND SNIPPETS Part 3

(Concatenate this URL with any filename below to run a video)

Chapter	Lecture	Snippet	Topic	http://www.designofmachinery.com/DOM/	Run Time
11	**24**		**Dynamic Force Analysis**	**Dynamic_Force_Analysis.mp4**	27:28
11		24A	*Single Link in Rotation*	*Single_Link_in_Rotation.mp4*	15:29
11		24B	*Fourbar Force Analysis*	*Fourbar_Force_Analysis.mp4*	12:18
11	**Lab**		**Fourbar Virtual Laboratory Video**	**Fourbar_Machine_Virtual_Laboratory.mp4**	35:38
11	**Lab**		**Fourbar Virtual Laboratory Data**	**Fourbar_Virtual_Lab.zip**	–
11	**25**		**Virtual Work and Flywheels**	**Virtual Work_and_Flywheels.mp4**	34:50
11		25A	*Virtual Work*	*Virtual_Work.mp4*	10:52
11		25B	*Flywheels*	*Flywheels.mp4*	24:07
12	**26**		**Balancing**	**Balancing.mp4**	48:09
12		26A	*Static Balance*	*Static_Balance.mp4*	09:58
12		26B	*Dynamic Balance*	*Dynamic_Balance.mp4*	09:42
12		26C	*Linkage Balancing*	*Linkage_Balancing.mp4*	26:55
12		26D	*Field Balancing*	*Field_Balancing.mp4*	02:43
12	**Lab**		**Fourbar Virtual Laboratory Video**	**Fourbar_Machine_Virtual_Laboratory.mp4**	35:38
12	**Lab**		**Fourbar Virtual Laboratory Data**	**Fourbar_Virtual_Lab.zip**	–
13	**27**		**Engine Kinematics**	**Engine_Kinematics.mp4**	48:17
13	**28**		**Engine Dynamics**	**Engine Dynamics.mp4**	53:17
13	**29**		**Engine Balancing and Pin Forces**	**Engine_Balancing_and_Pin_Forces.mp4**	42:38
13		29A	*Pin Forces*	*Pin_Forces.mp4*	20:03
13		29B	*Dynamic Balance*	*Balancing_One_Cylinder.mp4*	21:46
14	**30**		**Multicylinder Engines**	**Multicylinder_Engines.mp4**	44:25
14	**31**		**Even Firing**	**Even_Firing.mp4**	47:29
14	**32**		**Vee Engines**	**Vee_Engines.mp4**	48:25
14	**33**		**Balancing Multicylinders**	**Balancing_Multicylinders.mp4**	31:12
15	**34**		**Cam Dynamics**	**Cam_Dynamics.mp4**	48:29

DEMONSTRATION VIDEOS

(Concatenate this URL with any filename below to run a video)

Topic	http://www.designofmachinery.com/DOM/	Run Time
Boot Testing Machine	**Boot_Tester.mp4**	19:02
Bottle Printing Machine	**Bottle_Printing_Machine.mp4**	09:38
Grashof Condition	**The_Grashof_Condition.mp4**	24:12
High-Speed Spring Failure	**Spring_Failure.mp4**	03:46
Pick and Place Mechanism	**Pick_and_Place_Mechanism.mp4**	38:35
Spring Manufacturing Machinery	**Spring_Manufacturing.mp4**	12:23
Vibration Testing	**Vibration_Testing.mp4**	05:51

Note that you can download a PDF file containing hyperlinks to all the video content listed in the above tables. This allows print-book readers to easily access the videos without having to type in each URL as noted in the tables. Download the file:

http://www.designofmachinery.com/DOM/Video_Links_for_DOM_6ed.pdf